



PYTHIN FUNDAMENTALS

PROJECT: LOG ANALYZER

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1. Introduction

This report details the creation and implementation of the Python Fundamentals project: an advanced Log Analyzer script engineered to parse and analyze the Linux `/var/log/auth.log` file. By systematically extracting command usage patterns and vigilantly monitoring user authentication changes, the script fully complies with the outlined requirements, delivering critical insights into system security dynamics and potential vulnerabilities.

2. Tools and Commands Used

- I used Python 3 on my Ubuntu laptop for the script.
- Modules: `re` for matching patterns in logs, `sys` for handling errors.
- To edit: `nano log_analyzer.py`
- To run: `sudo python3 log_analyzer.py` (needs `sudo` for `auth.log`)
- Tested by running commands like `sudo ls`, `su`, `adduser` to make log entries.
- No extra stuff installed, just basic Python.

3. Log Analysis Output

3.1 . Extract Command Usage

Successful sudo commands with timestamp, user, and full command:

```
[2026-01-08 10:52:40] SUDO SUCCESS → User: peng-seyha | Command: /usr/bin/python3 TCI-2510-CAMBODIA-II.s20.xe105.py
[2026-01-08 10:52:40] SU USAGE → peng-seyha switched to root
[2026-01-08 10:54:17] SUDO SUCCESS → User: peng-seyha | Command: /usr/bin/python3 TCI-2510-CAMBODIA-II.s20.xe105.py
[2026-01-08 10:54:17] SU USAGE → peng-seyha switched to root
[2026-01-08 11:08:30] SUDO SUCCESS → User: peng-seyha | Command: /usr/bin/ls
[2026-01-08 11:08:30] SU USAGE → peng-seyha switched to root
[2026-01-08 11:08:42] SUDO SUCCESS → User: peng-seyha | Command: /usr/bin/python3 TCI-2510-CAMBODIA-II.s20.xe105.py
[2026-01-08 11:08:42] SU USAGE → peng-seyha switched to root
[2026-01-08 11:09:19] SUDO SUCCESS → User: peng-seyha | Command: /usr/bin/python3 TCI-2510-CAMBODIA-II.s20.xe105.py
[2026-01-08 11:09:19] SU USAGE → peng-seyha switched to root
```

3.1.2. Include the Executing User

Who executed the command (the username that ran sudo):

```
[2026-01-08 10:52:40] SUDO SUCCESS → User: peng-seyha | Command: /usr/bin/python3 TCI-2510-CAMBODIA-II.s20.xe105.py
[2026-01-08 10:52:40] SU USAGE → peng-seyha switched to root
[2026-01-08 10:54:17] SUDO SUCCESS → User: peng-seyha | Command: /usr/bin/python3 TCI-2510-CAMBODIA-II.s20.xe105.py
[2026-01-08 10:54:17] SU USAGE → peng-seyha switched to root
[2026-01-08 11:08:30] SUDO SUCCESS → User: peng-seyha | Command: /usr/bin/ls
[2026-01-08 11:08:30] SU USAGE → peng-seyha switched to root
[2026-01-08 11:08:42] SUDO SUCCESS → User: peng-seyha | Command: /usr/bin/python3 TCI-2510-CAMBODIA-II.s20.xe105.py
[2026-01-08 11:08:42] SU USAGE → peng-seyha switched to root
[2026-01-08 11:09:19] SUDO SUCCESS → User: peng-seyha | Command: /usr/bin/python3 TCI-2510-CAMBODIA-II.s20.xe105.py
[2026-01-08 11:09:19] SU USAGE → peng-seyha switched to root
```

3.1.3. Include the Command

Every SUDO SUCCESS and ALERT! FAILED SUDO line shows the full command.

```
[2026-01-08 10:47:25] SU USAGE → peng-seyha switched to root
[2026-01-08 10:48:40] ALERT! FAILED AUTHENTICATION → pam_unix(sudo:auth): authentication failure; logname=peng-seyha
uid=1000 euid=0 tty=/dev/pts/3 ruser=peng-seyha rhost= user=peng-seyha
[2026-01-08 10:48:49] ALERT! FAILED SUDO → User: peng-seyha | Attempted: /usr/bin/ls
```

3.2. Monitor User Authentication Changes

The script continuously monitors /var/log/auth.log and prints clear details for all authentication-related events, including timestamps.

```
[2026-01-08 10:47:25] SU USAGE → peng-seyha switched to root
[2026-01-08 11:08:30] SUDO SUCCESS → User: peng-seyha | Command: /usr/bin/ls
[2026-01-08 11:08:30] SU USAGE → peng-seyha switched to root
[2026-01-08 11:08:42] SUDO SUCCESS → User: peng-seyha | Command: /usr/bin/python3 TCI-2510-CAMBODIA-II.s20.xe105.py
[2026-01-08 11:08:42] SU USAGE → peng-seyha switched to root
[2026-01-08 11:09:19] SUDO SUCCESS → User: peng-seyha | Command: /usr/bin/python3 TCI-2510-CAMBODIA-II.s20.xe105.py
[2026-01-08 11:09:19] SU USAGE → peng-seyha switched to root
```

3.2.1. Print Details of Newly Added Users, Including the Timestamp

Detects user creation (adduser). Prints timestamp and full user details (name, UID, GID, home directory, shell).

```
[2026-01-08 11:23:24] SU USAGE → peng-seyha switched to root
[2026-01-08 11:23:32] SUDO SUCCESS → User: peng-seyha | Command: /usr/sbin/adduser demo-user
[2026-01-08 11:23:32] SU USAGE → peng-seyha switched to root
[2026-01-08 11:23:32] NEW USER ADDED → new user: name=demo-user, UID=1006, GID=1006, home=/home/demo-user, shell=/bin/bash, from=/dev/pts/3
```

3.2.2. Print Details of Deleted Users, Including the Timestamp

Detects user deletion (deluser). Prints timestamp and deleted username.

```
[2026-01-08 11:27:02] SU USAGE → peng-seyha switched to root
[2026-01-08 11:27:02] USER DELETED → delete user 'demo-user'
[2026-01-08 11:27:06] SUDO SUCCESS → User: peng-seyha | Command: /usr/bin/python3 TCI-2510-CAMBODIA-II.s20.xe105.py
[2026-01-08 11:27:06] SU USAGE → peng-seyha switched to root
```

3.2.3. Print Details of Changing Passwords, Including the Timestamp

Detects password changes (passwd or during user creation). Prints timestamp and affected user.

```
v/pts/3
[2026-01-08 11:23:38] PASSWORD CHANGED → User: demo-user
[2026-01-08 11:23:49] SUDO SUCCESS → User: peng-seyha | Command: /usr/bin/python3 TCI-2510-CAMBODIA-II.s20.xe105.py
[2026-01-08 11:23:49] SU USAGE → peng-seyha switched to root
```

3.2.4. Print Details of When Users Used the su Command

Detects successful su usage. Prints timestamp, source user, and target user.

```
[2026-01-08 11:27:06] SU USAGE → peng-seyha switched to root
```

3.2.5. Print Details of Users Who Used the sudo; Include the Command

Detects successful sudo commands. Prints timestamp, executing user, and full

```
[2026-01-08 10:52:40] SUDO SUCCESS → User: peng-seyha | Command: /usr/bin/python3 TCI-2510-CAMBODIA-II.s20.xe105.py
[2026-01-08 10:52:40] SU USAGE → peng-seyha switched to root
[2026-01-08 10:54:17] SUDO SUCCESS → User: peng-seyha | Command: /usr/bin/python3 TCI-2510-CAMBODIA-II.s20.xe105.py
[2026-01-08 10:54:17] SU USAGE → peng-seyha switched to root
[2026-01-08 11:08:30] SUDO SUCCESS → User: peng-seyha | Command: /usr/bin/ls
[2026-01-08 11:08:30] SU USAGE → peng-seyha switched to root
[2026-01-08 11:08:42] SUDO SUCCESS → User: peng-seyha | Command: /usr/bin/python3 TCI-2510-CAMBODIA-II.s20.xe105.py
[2026-01-08 11:08:42] SU USAGE → peng-seyha switched to root
[2026-01-08 11:09:19] SUDO SUCCESS → User: peng-seyha | Command: /usr/bin/python3 TCI-2510-CAMBODIA-II.s20.xe105.py
[2026-01-08 11:09:19] SU USAGE → peng-seyha switched to root
```

3.2.6. Print ALERT! If Users Failed to Use the sudo Command

Detects failed sudo authentication (wrong password). Prints ALERT! with timestamp, user, and attempted command.

```
[2026-01-08 11:27:00] SU USAGE → peng-seyha switched to root
[2026-01-08 11:32:48] ALERT! FAILED AUTHENTICATION → pam_unix(sudo:auth): authentication failure; logname=peng-seyha uid=1000 euid=0
tty=/dev/pts/2 ruser=peng-seyha rhost= user=peng-seyha
[2026-01-08 11:32:58] ALERT! FAILED SUDO → User: peng-seyha | Attempted: /usr/bin/ls
```

4. Conclusion

The Log Analyzer project has been successfully completed using Python to parse and analyze the /var/log/auth.log file on Kali Linux. The script effectively fulfills all required objectives by extracting and displaying critical security events with clear timestamps and details.